

L11: The Optimal Search Exponent in Two Dimensions

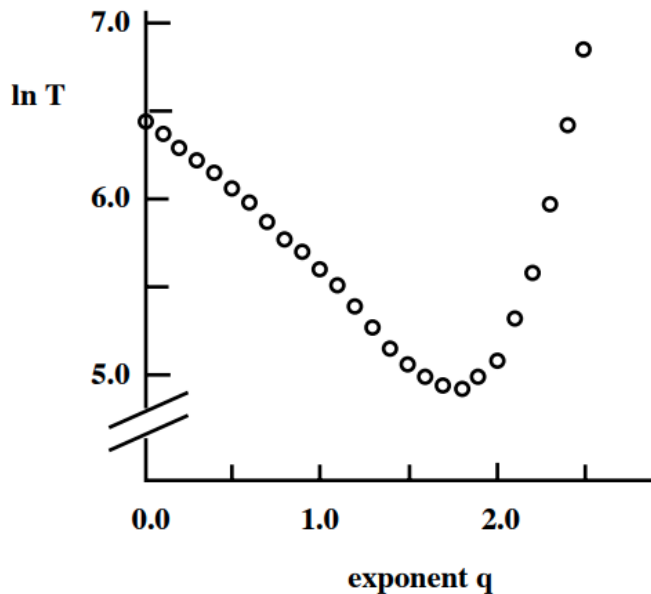


Image Source: Textbook [“Networks, Crowds and Markets”](https://www.cs.cornell.edu/home/kleinber/networks-book) by Easley and Kleinberg. Figure 20.6

Let us start with some simulation results. The plot shows the exponent q at the x-axis, when the initial grid has $k=2$ dimensions. The y-axis shows the *logarithm of the expected delivery time* in a grid with 400 million nodes. Each point is an average across 1000 runs with different source-target pairs and different assignments of random edges.

As expected, values of q that are either close to 0 or quite large result in very slow delivery times. The random edges are either *“too random”* (when q is close to 0) or *“not random enough”* (when q is large).

It appears that **the optimal value of the exponent q is when it is close to 2** – which, remember, is also the value

of the dimension k in these simulations. Is this a coincidence?

The answer is that we can prove that the optimal value of q is equal to k . In other words, if we want to maximize the speed of a decentralized search in a k -dimensional world, we should have random *“shortcut links”* the probability of which decays as $d(v, w)^{-k}$ with the distance between two nodes. This surprising result was proven by Kleinberg in 2000.

You can find the proof, at least for $k=1$ and 2, in the online textbook *“Networks, Crowds and Markets”* by Easley and Kleinberg (section 20.7). On the next page, we give the basic intuition behind this result.

Food For Thought

Why do you think that the average delivery time increases much faster when $q > 2$ than when $q < 2$?